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Project: A16730S1

X-ray single photon energy dispersive spectroscopy

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Abstract

The emission spectrum of an X-ray tube with a germanium target is obtained through analysis of images taken with a charge coupled device of x-rays scattered off a beryl crystal. Single photon counting, developed using synthetic data, and Bragg's law were used to obtain a spectrum of energies between 1095 and 1638 eV which highlighted germanium's L series emission lines as well as bremsstrahlung.

Introduction

In some spectroscopy experiments, two-dimensional data can be acquired by use of a charge-coupled device (CCD) onto which, diffracted radiation falls. In this investigation a Princeton Instruments PI-MTE 2048B CCD camera was used, consisting of 2048 by 2048 square pixels of side length 13.5 μm [1]. A beryl crystal was present to diffract radiation emitted by an x-ray tube with a germanium target onto the CCD which recorded 11 sets of data. The CCD was aligned to record incident x-ray photons in the energy range found to be 1095-1638 eV; since properties of the beryl diffraction crystal are known, we can calculate the range of glancing angles of diffracted x-ray photons that are incident on the CCD using Bragg's law,

$$n\lambda = 2d \sin \vartheta. \quad (1)$$

Here, n is diffraction order; λ is radiation wavelength; d is the grating constant and ϑ is the glancing angle of incident radiation. The data provided was obtained by first order diffraction off a family of planes in the beryl crystal with spacing $2d = 1.5954 \text{ nm}$. Using *equation 1* we obtain the range of glancing angles 28.3-45.2 $^\circ$.

The aim of the project is to return a spectrum of the Ge x-ray source by using single photon counting and Bragg's law when analysing spectroscopy data from the CCD. A single set of data is a two-dimensional array of integer values; each value is provided by a single pixel in the CCD and is referred to as the ADU value. Single photon counting is employed to use the ADU values to specify the location of photon events on the CCD. Bragg's law is then used to obtain direct photon energy information as a function of position of photon events. The resulting spectrum is a collection of these photon energies.

I will begin by discussing properties of the provided data sets after which I investigate the methods and results of the project's sub-sections. These sections are, fitting the $L\alpha/L\beta$ emission lines, obtaining photon energy as a function of CCD position, creating synthetic data, and developing a single photon counting algorithm. Finally, the x-ray emission spectrums will be analysed, and conclusions drawn.

Data

As discussed in the introduction, the data is from a CCD camera and is in the form of sets of 2048 by 2048 square arrays of integer ADU values. The ADU value is a measure of the excitation of a pixel in the CCD. A photon incident on the CCD will cause excitation and result in increased ADU value in pixels within its vicinity. CCDs are sensitive and will produce signal in the dark due to factors including electronic readout, thermal excitement, and shot noise. This noise can be identified via the

gaussian ‘pedestal’ as seen on the left of *Figure 1*. 20 data sets were provided, 9 of these were dark-images and 11 contained photon events.

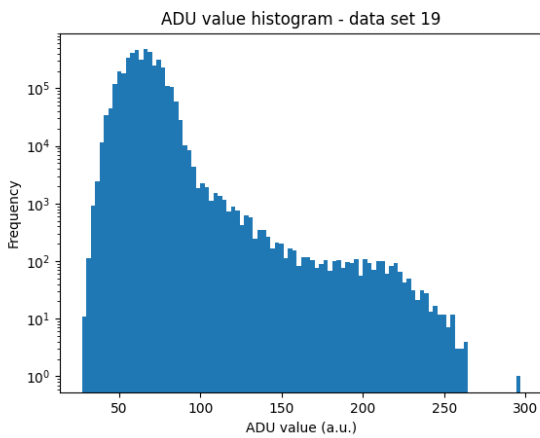


Figure 1: Histogram of ADU values from a non-dark image (data set 19)

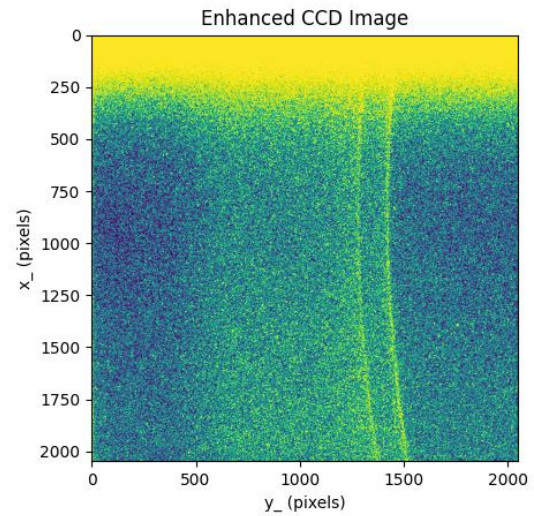


Figure 2: Enhanced CCD image of combined non-dark data sets; each pixel is a square with side length $13.5 \mu\text{m}$

In *Figure 1*, the pedestal can be seen clearly along with higher ADU values due to photon events. Most pixels have their value within the pedestal as expected; roughly $10^4 - 10^5$ photon events are expected within each data set which is far below the total pixel count of approximately 4×10^6 .

Figure 2 highlights another key feature of the data: the two $L\alpha/L\beta$ emission lines of the germanium spectrum. These peaks occur at x-ray energies of 1188.0 and 1218.5 eV [2] which means along these lines, x-rays have well defined energy due to the sharpness of these spectral peaks (the lines appear broad in *Figure 2* only due to the applied image enhancement techniques). *Equation 1* tells us that a well-defined photon energy corresponds to a well-defined glancing angle and using the emission line energies along with beryl’s grating constant from before, we obtain glancing angles of 40.856° and 39.626° respectively for the emission lines. Curve fitting these emission lines will allow us to understand the geometry of the experimental arrangement and the shape of photon energy space on the CCD. From *Figure 2* we can also deduce that the direction of energy dispersion is roughly along the y_- -axis due to the orientation of the emission lines.

Emission Line Fitting – Method and Results

Since the diffraction angles of the two emission lines are known, we can use their curvature along with the distance between them to model the geometry of the system and assign a glancing diffraction angle to every pixel in the CCD. Since diffraction angle and energy are directly related by *Equation 1*, we can assign an energy value to photons arriving at any point on the CCD and hence build an emission spectrum.

The first step is to locate the pixels that fall under an emission line. I took all the non-dark data sets provided and summed the ADU values of each pixel to obtain a single set. Photon events are generally sparse across the CCD (there are many more pixels than photon events) but are expectedly much denser along emission lines resulting in an increased on-line signal to noise ratio in the summed data set. This allowed the lines to be found easily by taking local sums of pixels across our new data set. With our data orientated as see in *Figure 2* the array was divided into rows of equal thickness; rows from 2 to 128 pixels thick were investigated in steps of powers of 2. For each row, ADU value sums of small groups of adjacent pixels were taken and the location of the maximum sum produced was returned. The emission line regions should have the largest ADU sums as they have the most photon events. To determine how many pixels to take for the small sums, a variety of rectangles with varying width were investigated.

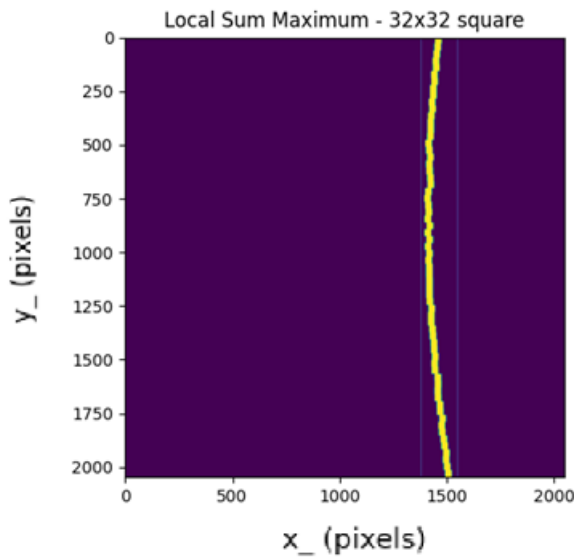


Figure 3: 1188.0 eV emission line located with maxima of local sums

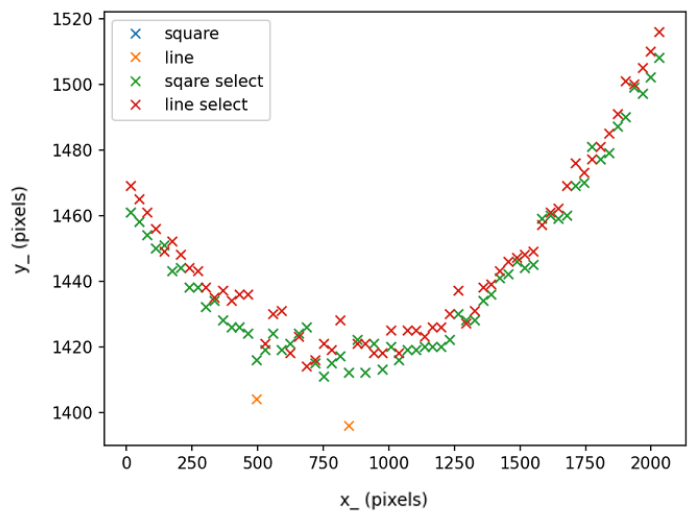


Figure 4: Location of maximum local sums, with anomalies selected, for square and line summation shapes

Figure 3 shows the example of local ADU sums being taken across rows 32 pixels wide. In each row, sums of all possible 32x32 pixel squares were taken and *Figure 3* shows the squares of maximum ADU sum highlighted in yellow. The lines on either side of the squares are boundaries to take the sums within for computational efficiency and to separate the two emission lines. I

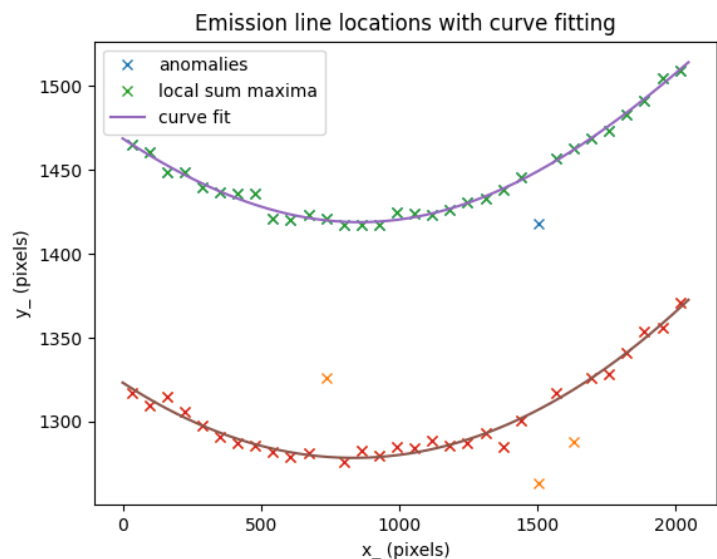


Figure 5: Curve fitted emission lines with anomalies selected

determined the best shape for this endeavour was to take sums of vertical lines (1-pixel thick rectangles) as opposed to squares. *Figure 4* displays the difference between square and line sums; there is a clear displacement between the two and I found visually, the points obtained using line-sums clustered closer to the emission lines. Rows of 64-pixel width were chosen to balance having enough pixels to return a meaningful sum but few enough to obtain enough data points to fit the emission lines.

The above method was used to gather 32 equally spaced location points across each of the two emission lines. These had to be cleaned by removing anomalies that were considerably displaced from their neighbours as seen in *Figure 5*. The two sets of points were then separately modelled using SciPy's [3] least squares curve fitting algorithm with the below equation.

$$y_- = a + b \cdot \sqrt{c + d \cdot x_-^2 + e \cdot x_-} + f \cdot x_- \quad (2)$$

Equation 2 is a modified form of a general conic section equation and has a, b, c, d, e, f as constants to be determined. The x_- and y_- in *Equation 2* are in units of CCD pixels and follow the directions shown in *Figure 5* (along the edges of the CCD). When executing the least squares fitting algorithm, runtime timeouts were common. To reduce this phenomenon, I included the seemingly superfluous constant ' b ', this relaxed the range of initial inputs for the parameters that would return a result. I also ran fitting on a modified *Equation 2* which had the $f \cdot x$ term removed to reduce timeouts after which a fit with the full equation proceeded. This worked well as the ' f ' constant was always small, so the results produced from the modified equation worked as initial inputs for the parameters of the fit using the full *Equation 2*.

A result that will be important later is the distance between the spectral lines, Δy . My fits obtained the output $\Delta y = 142$ pixels.

Geometry – Energy as a Function of Position - Method

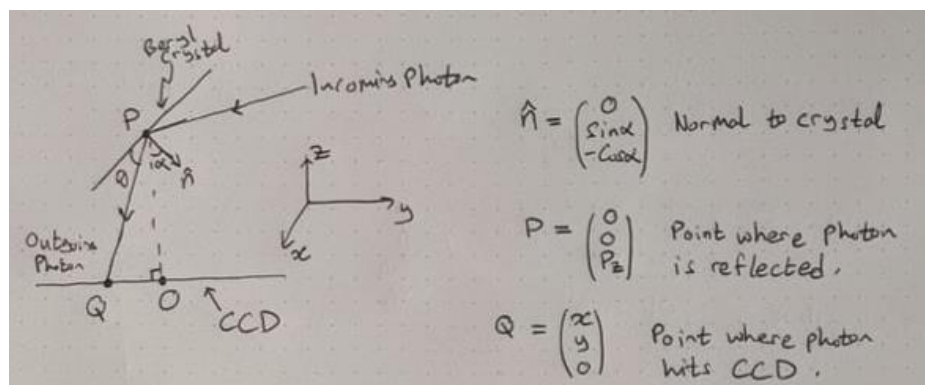


Figure 6: Model diagram of the experimental arrangement to obtain CCD images [4]

Figure 6 shows a model of the experimental arrangement with the CCD defining the x, y plane and with the y -axis defined as the direction of energy dispersion. Interesting, unknown parameters include α , the angle between the normal to the beryl crystal's diffraction plane and the normal to the CCD, and $|\overline{OP}|$ which we will refer to as P , the distance normal to the CCD between the point where a photon diffracts off the crystal and the CCD itself. It follows from *Figure 6* that:

$$\overline{PQ} \cdot \hat{n} = |\overline{PQ}| \cos(90 - \vartheta), \quad (3)$$

$$\Rightarrow y \sin \alpha + P \cos \alpha = \sqrt{x^2 + y^2 + P^2} \cdot \sin \vartheta, \quad (4)$$

where $\vec{n}$ is the beryl crystal's unit normal vector, ϑ remains the glancing diffraction angle, and the other symbols are defined above.

We can also deduce from *Figure 6* that

$$\Delta y = P \left(\frac{1}{\tan(\vartheta + \alpha)} - \frac{1}{\tan(\vartheta' + \alpha)} \right) \quad (5)$$

where Δy is the y-axis separation between two photons incident to the CCD with glancing diffraction angles ϑ and ϑ' .

Equation 5 can be used to obtain a relationship between α and P when we take ϑ and ϑ' to be the known diffraction angles of the two L series emission lines and Δy to be simply the distance, along the axis of the direction of energy propagation, between the two emission lines which we can obtain from our curve fits of these maxima. When considering the direction of energy propagation, I reviewed my fitted conic section equations for the emission lines. From *Equation 2*, the ' f ' constant would be zero if the propagation direction were along the y_- -axis. This would mean the fitted emission lines have a line of symmetry parallel to the y_- -axis. Since it is impossible to perfectly align the apparatus when taking data, we expect the propagation direction to be close to, but not perfectly parallel with the y_- -axis; this is confirmed with a non-zero value for ' f '. To calculate the angle ϕ , between the direction of energy propagation and the CCD's y-axis (as seen in *Figure 5*) I derived the following equation,

$$\phi = \frac{1}{2} \tan^{-1} \left(\frac{2f}{d \cdot b^2 - f^2} \right) \quad (6)$$

where f , d and b are as they appear in *Equation 2*.

Equation 4 will be used to assign every pixel in the CCD a value for ϑ , the expected glancing diffraction angle of a photon which lands on that pixel. To use *Equation 4* in this way, we need to convert from our x_- , y_- coordinates that run parallel to the edges of the CCD, into the x , y coordinates that run perpendicular and parallel to the direction of energy propagation respectively. Both bases lie within the plane of the CCD. We can use *Equation 6* to obtain the rotation between bases and then finally we must find the location of the new origin. We know $x = 0$ occurs at any point on the y_- axis which is simply the line of symmetry bisecting both our emission lines. We can see geometrically from studying *Figure 6* that $y = 0$ occurs a distance $\frac{P}{\tan(\vartheta + \alpha)}$, along the energy propagation direction y , from the point on an emission line where $x = 0$, in the case that ϑ is the glancing diffraction angle for that specific emission line.

To use *Equation 4*, we need definite values for P , and α . As mentioned above, *Equation 5* can be used to write P as a function of α , so we need only select a numerical value for α . My method for selecting this value was to model an emission line by using *Equation 4*. I modelled the line with various values of α and chose P by using *Equation 5* with my value of Δy obtained from fitting the data. This model line was then compared to the curve obtained from the least square fitting of emission lines in the data. I applied a least square fitting technique, again using SciPy [3], to return the value of α that produced the model emission line which most closely fit the emission line

obtained from the data. This method could be repeated and used on both two emission lines to return two values of α to be compared.

Once we have chosen α , we can use *Equation 5* to obtain P and then using the above method, convert our known x_-, y_- CCD pixel coordinates into x, y energy propagation coordinates first seen in *Figure 6*. These constants and coordinates will then spatially describe the CCD and turn *Equation 4* into a function of only one unknown variable, ϑ , the glancing diffraction angle of photons. Hence, we can calculate and assign each pixel in the CCD a value of ϑ based on the pixels position. Using Bragg's law, *Equation 1*, these assigned ϑ values can be converted to photon energy values, and we obtain the 2048 by 2048 energy matrix.

Geometry – Energy as a Function of Position – Results

The rotation angle ϕ between our x_-, y_- and x, y coordinates, obtained using *Equation 6* gave a value between 0.507° and -0.541° when calculated for either spectral line, fitted using various forms of *Equation 2*. These angles are significantly small and average to roughly zero, so I chose this as my value of ϕ .

For the value of α , the least squares fitting method yielded the value $\alpha = 1.043$ radians when used on the 1188.0 eV emission line and $\alpha = 1.045$ radians on the 1218.5 eV line. These values agree well, and a mean $\alpha = 1.044$ radians was chosen to calculate the energy matrix.

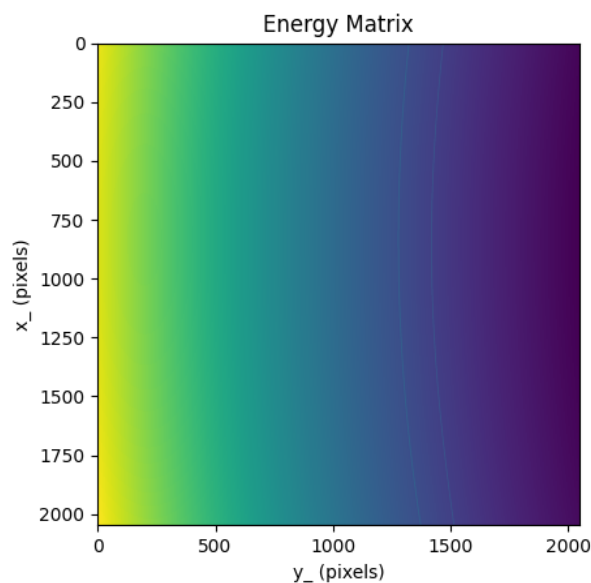


Figure 7: Matrix of CCD pixels with corresponding photon energy values

Figure 7 shows the calculated energy matrix which represents the CCD and assigns a photon energy value in eV to each pixel. The two highlighted lines in *Figure 7* are not jumps in energy, but lines of constant energy (1218.5 and 1188.0 eV) highlighted to demonstrate the curvature of the space. The range of energies present is 1079 to 1667 eV however, lines of constant energy at the extremes get cut off at the edges of the matrix due to its square shape. Hence photon events at the edges of this range are rare as they correspond to only a few pixels. The range of energies that do not get cut off is 1095 to 1638 eV.

Synthetic Data

I endeavoured to create synthetic CCD data on which to test my single photon counting (SPC) algorithm. This was useful as I would know the exact location of my synthetic photons to compare to the locations my SPC algorithm returns. I began by modelling the germanium x-ray emission spectrum with a bremsstrahlung ‘breaking radiation’ curve using an equation for the energy dependent, free-free emissivity [5] you will find in the appendix. On top of this, I summed two sharp gaussian peaks to model the L series resonances. I chose the standard deviation for the width of these maxima as 1.5 eV which I calculated from the full width half maximum of Ge L series emission [6].

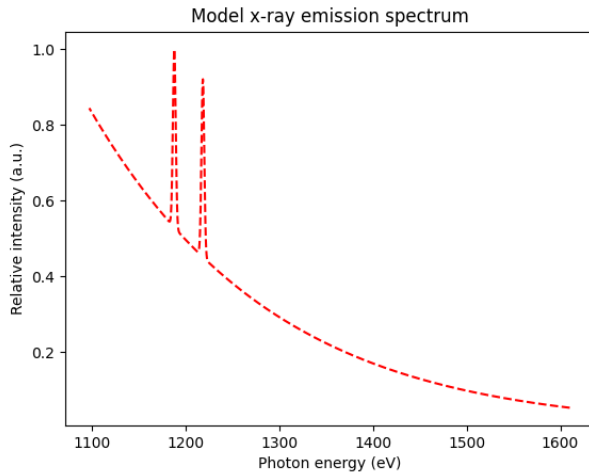


Figure 8: Model germanium X-ray tube emission spectrum

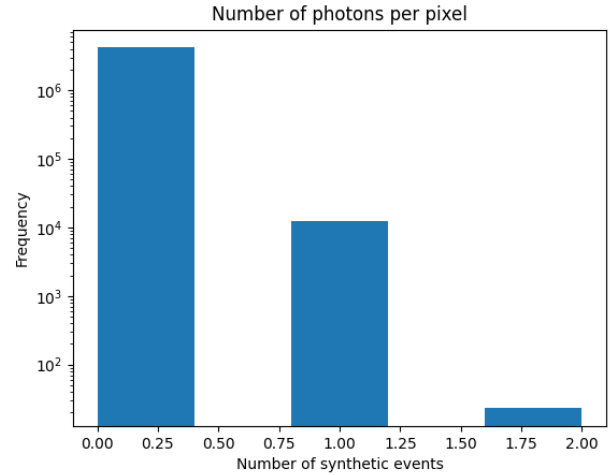


Figure 9: Histogram of model photon number per pixel

My model x-ray emission spectrum shown in *Figure 8* has relative intensity in arbitrary units on the y-axis; I normalised this spectrum and used the resulting distribution as a probability function to determine the likelihood of photon events of a specific energy occurring. When assigning an energy value to each pixel in my synthetic data, I decided to make each column in my 2048 by 2048 array a line of constant energy. The first column started at 1097 eV and subsequent columns increased linearly in energy up to a maximum value of 1610 eV. This was a good simplification as my synthetic data was created only to develop my SPC algorithm which involves identifying photon events and is not affected by the geometry of the energy dispersion. Having assigned energies to my pixels I could use the previously discussed probability function to assign an expected value for the number of photon events occurring at each separate pixel and modelled each pixel with its own poisson distribution. When normalising the probability function, I reduced its magnitude further to reduce the total number of photon events I would produce. I aimed for the total number of synthetic photon events to be in the $10^4 - 10^5$ range. *Figure 9* shows the breakdown of these event numbers; as expected, most pixels do not experience a photon event, and most of those that do, experience only 1 photon.

To simulate the ADU value of a photon event, I decided on a gaussian distribution for the total ADU contribution of a single photon. This total ADU would then be shared probabilistically between the event pixel and its neighbours. I chose the mean as 160 and the standard deviation as 30 for this total ADU distribution due to the mean and FMHW of the below histogram.

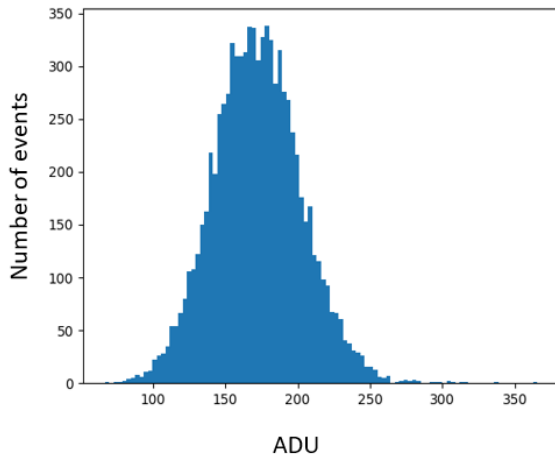


Figure 10: Histogram of pedestal subtracted total ADU of single photon events from data set 6

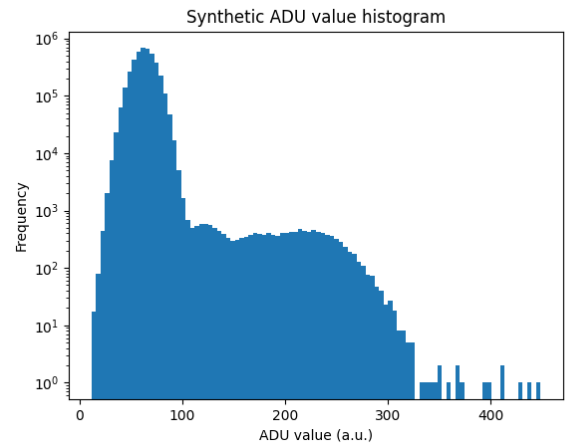


Figure 11: Histogram of ADU values from synthetic data

Figure 10 shows a histogram of the sum of ADU values in a 1-pixel radius of a real single photon event. The pedestal mean has been subtracted so the total ADU plotted is that above the gaussian noise. To select real single photon events for Figure 10, points are taken from data set 6 that exceed a major threshold of 120 but whose adjacent pixels all reside within 3 standard deviations of the pedestal. Taking a sum over 9 pixels reduces the effect of natural variance in the pedestal.

To determine how many pixels a synthetic photon spreads into, I used a poisson distribution as it agreed well with results from the real data as seen in the table below.

	Number of adjacent pixels that a photon event spreads into			
	0	1	2	3
Real data	0.667	0.261	0.051	0.017
Poisson $\lambda=0.4$	0.670	0.268	0.054	0.008

Table 1: Probability of pixel spread from data set 6 compared to a poisson distribution

Data was selected in a similar manner to that of data shown in Figure 10 and probabilities obtained by dividing the number of events spreading to a specific number of pixels by the total number of selected events.

To determine what fraction of the total ADU went to each pixel, once the pixel spread number was assigned, I used a probabilistic model parameterised by a mean spread factor of 0.64. This was obtained by comparing the total ADU as seen in Figure 10 to the mean value of the major pixel in each of these events.

Figure 11 shows the histogram of ADU values produced in my synthetic data. There is a discrepancy between the number of counts in the 100 to 150 ADU range in the synthetic data when compared with real data. This is due to charge spreading being present in most real photon events which is neglected in synthetic data creating for the case where spreading is small.

Single Photon Counting

My SPC algorithm not only has to locate photon events but remove erroneous data points. In all test data sets provided, a bright spot on the CCD appeared at and around the [1396, 1758] pixel. To remove this, as well as any other anomalously large ADU regions, I added a large threshold for the ADU sum of a pixel, and its 8 neighbours. Exceeding this results in the points not being counted as photon events.

The most effective method I found for counting single photons involved thresholding. I chose a major threshold of $\text{ADU} = 120$. Pixels exceeding this value would be selected as major pixels, and the square of points around these (2 or fewer pixels away from the major pixel) investigated. The 8 adjacent pixels are run past a second, minor threshold which is defined as the mean of the data set's pedestal plus 3 standard deviations. The mean and standard deviation of the gaussian pedestal for a single set are extracted using an algorithm I wrote. Through all the provided data sets, the mean of the pedestal ranged from 50 – 64 and standard deviation from 8.1 – 10.6. Hence the maximum minor threshold possible is 96. 120 for the major threshold is clearly large enough to avoid pedestal noise however, a larger value risks missing photon events; my employed 120 threshold only picks up about half the photon events I eventually count so increasing it would be unwise.

Major pixels whose neighbours all fall below the minor threshold can be identified as photon events with little uncertainty. The total ADU value can be examined to determine how many events are present; we expect about 160 above the pedestal mean for a single photon event. Photon events which spread into multiple pixels introduce the most uncertainty. These were dealt with by taking the major and surrounding pixels and calculating their total ADU above that expected from background noise. *Figure 10* shows the distribution of these totals which can be used to assign a photon number with uncertainty for each case by taking a mean of 160 and standard deviation 30 directly from the figure.

To capture remaining photon events with significant spread such that there is no major pixel, the above total ADU method was used on local regions of increased ADU. When no major pixel is present, greater care must be taken to avoid erroneously indicating a photon event is present. Hence, I increased the threshold that the calculated total ADU must exceed to register a photon event.

Emission Spectra

Single photon counting produced photon event number with uncertainty at all positions within the CCD. This was combined with the energy matrix to return a list of all the photon energies. The histogram of this list is the x-ray emission spectra.

Figures 12 and 13 resemble the expected spectrum shown in *Figure 14*. Similarities include spectral peaks at 1218.5 and 1188.0 eV which are sharp on the lower energy side and appear broader on the higher energy side. Also present is the bremsstrahlung emission curve which oscillates and appears less smooth as energy increases; this is highlighted particularly in *Figure 13*.

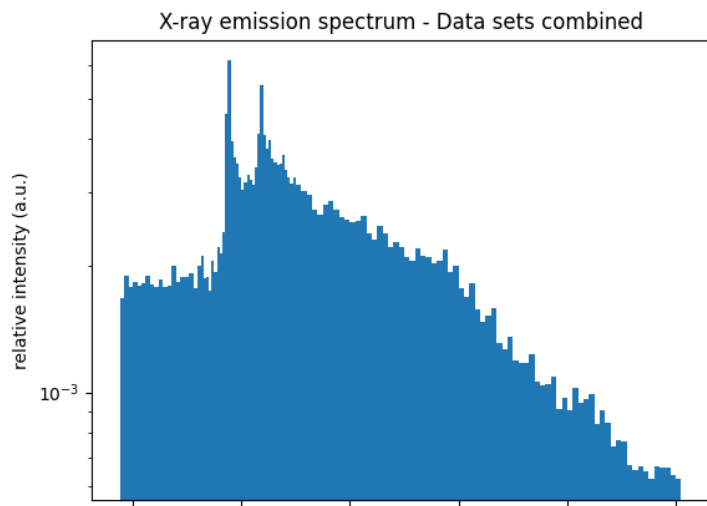


Figure 12: X-ray emission spectrum obtained from all non-dark data sets

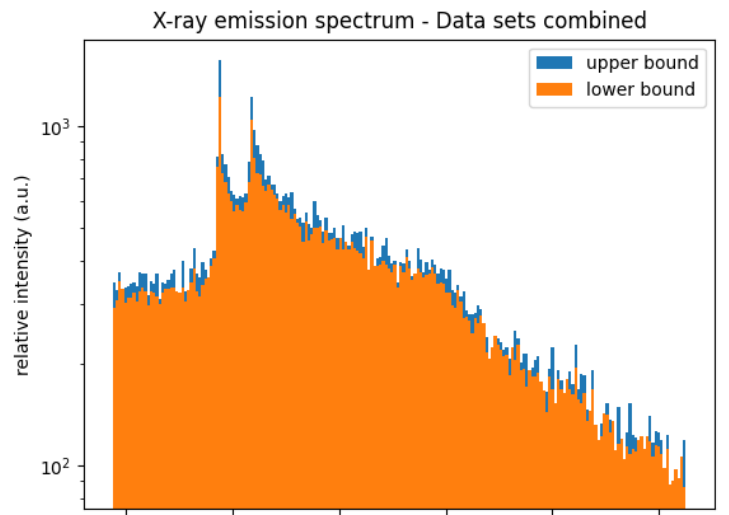


Figure 13: Uncertainty on the x-ray emission spectrum obtained from all non-dark data sets

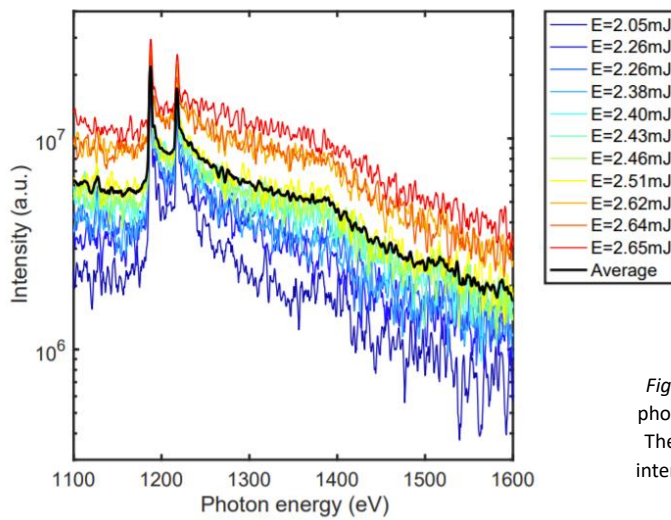


Figure 14: Spectra obtained for different shots with a XFEL photon energy of 1400 eV as a function of the beam energy. The small variations in the beam energy affect the spectral intensity, and effectively, the temperature of the plasma. [7]

Conclusions

I have created a software suite that analyses x-ray spectroscopy data collected from a Charge Coupled Device and returns a spectrum of the radiation source. Single photon counting was used to locate photon events and provide uncertainty on the number of photons. Since properties of the diffraction crystal were known, and since the geometry of the system could be modelled by locating the two L-series emission lines (at 1218.5 and 1188.0 eV), it was possible to extract energy information directly from Bragg's law.

Synthetic data was created using parameters chosen from analysis of real data sets. The synthetic data was used to test and optimize the photon counting algorithm. X-ray emission spectra obtained from multiple data sets were combined to give my final spectrum which agreed well with the expected result and included the notable features of sharp lines at exactly 1188.0 and 1218.5 eV and the continuous curve due to bremsstrahlung.

Further work should be done on creating synthetic data as there is a discrepancy between the number of counts in the 100 to 150 ADU range in the synthetic data when compared with real data. Introducing some small charge spreading in supposedly 'single pixel' synthetic photon events could help this discrepancy. Improvements would allow us to re-analyse the photon counting algorithm and adjustments could be made such as altering thresholds.

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Appendix

$$j_{\text{ff}}(\lambda) = \left(\frac{e^2}{4\pi\epsilon_0} \right)^3 n_e^2 \frac{16\pi Z e^{-2\pi\hbar c/\lambda k_B T_e}}{3m_e c^2 \lambda^2 \sqrt{6\pi k_B T_e m_e}} g_T(\lambda) \quad (1)$$

for the free-free emissivity $j_{\text{ff}}(\lambda)$. Here, m_e is the electron mass and $g_T(\lambda)$ is the wavelength-dependent Gaunt factor [5]